

OP Jindal Global University



Jindal Global Business School

MBA- Elective

Power BI for Managers

Report: JGBS-Course Structure Dashboard

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Introduction

In the dynamic landscape of education, Effective data management and analysis are essential to improving the entire academic experience in the ever-changing world of education. By utilizing data from the university administration office, this Power BI dashboard analysis seeks to give a thorough overview of the course structure inside JGU's Jindal Global Business School (JGBS). Giving professors, staff, and students useful information on how courses are structured and assigned is the main goal in order to enable stakeholders to make well-informed decisions and promote ongoing development. We will examine several important aspects of the Business School's course offerings as we read through this report, such as program distribution, course subjects, course credits, course distribution by semester, how courses are carried over to subsequent semesters, and other pertinent metrics. By harnessing the power of data visualization and analytics, this report seeks to shed light on patterns, trends, and potential areas for optimization within the academic curriculum.

JGBS has various business management programs which includes, BBA(HONS), BBA(BA), BBA(FM), BBA(FB), IBM, MBA(BA), MBA (ONLINE), MBA and GLOBAL MBA(BA).

By leveraging the capabilities of Power BI, we aim to present an interactive and user-friendly interface that allows users to explore the data seamlessly. Whether you are an academic advisor seeking to understand trends of program specific course availability or a student who want to understand the course availability in various program provided by JGBS, this report is designed to cater to a diverse audience and provide actionable insights by using Power BI.

The report will delve into major areas such as problem statements and the objective of creating a Power BI dashboard. Using snapshots, we'll showcase how various functions, tables, and cards are used for data analysis. Concluding with visual representation of data and its analysis, the report aims to provide a comprehensive overview of the Business School's course structure at JGU, and how data is converted into a visual representation providing all the information at a single dashboard.

The collaborative effort between the university administration office and the creators of this Power BI dashboard signifies a commitment to transparency, efficiency, and data-driven decision-making. As we embark on this journey through the Business School's course structure, let us unlock the potential of data to enhance the educational experience and contribute to the continuous advancement of academic programs at JGU.

Problem statement

The traditional use of Excel for managing our Business School's course data is challenging. The cluttered nature of Excel sheets limits our ability to analyze and interpret information effectively. It becomes difficult to make wise decisions based on the course data when there aren't enough options for varied visualization. This report aims to address these issues by leveraging Power BI

to create a more visually intuitive and user-friendly dashboard, enhancing the accessibility and interpretability of crucial academic information.

Objectives

This report aims to modernize the management and analysis of course data within the Jindal Global Business School (JGBS), using the dynamic capabilities of Power BI. The primary objective is to overcome challenges associated with conventional methods by providing stakeholders with a user-friendly dashboard, facilitating easy understanding and analysis through visually intuitive representations of program distribution, course subjects, credits, and semester-wise trends. The transition to Power BI seeks to empower faculty, administrators, and students, streamlining data analysis, facilitating informed decision-making, and contributing to the continuous improvement of academic programs at JGU.

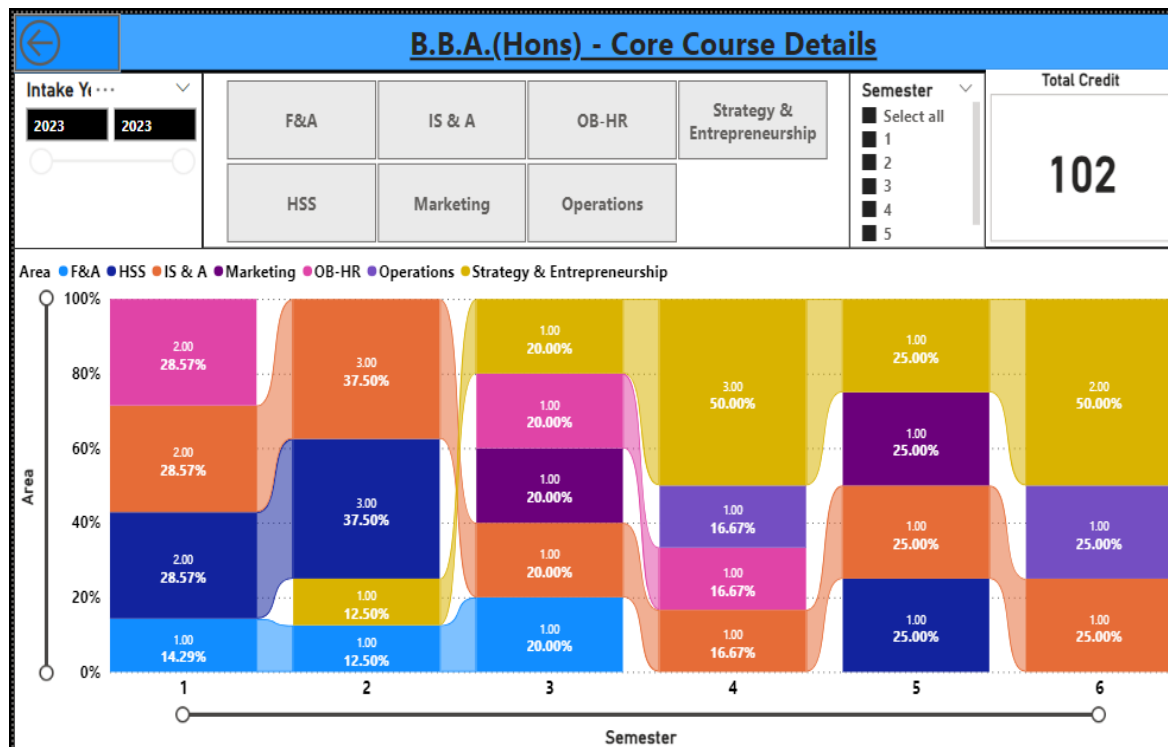
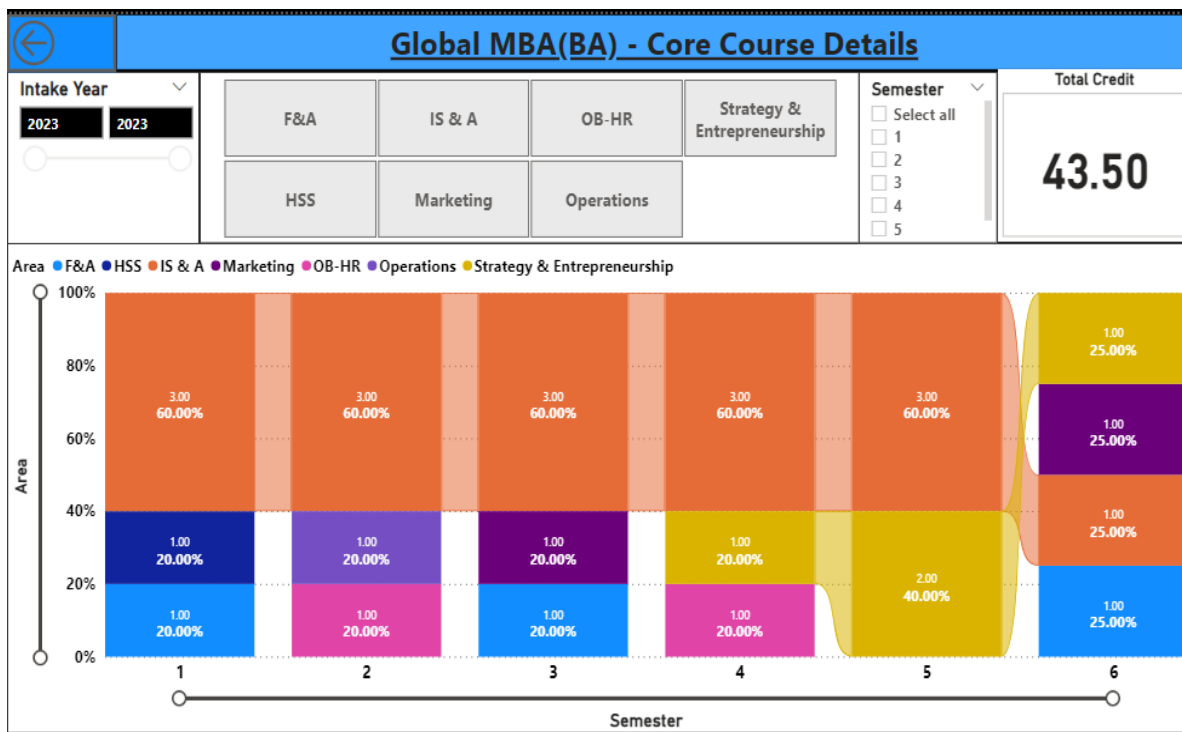
Data Collection

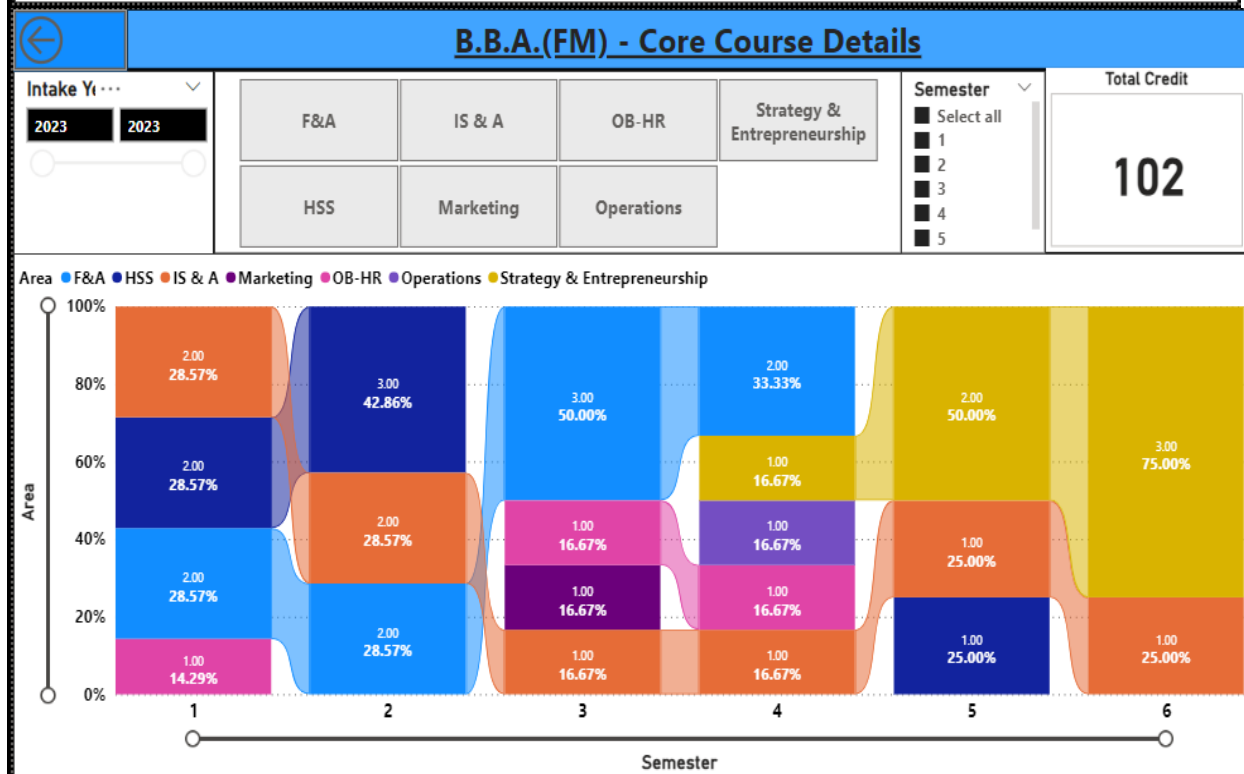
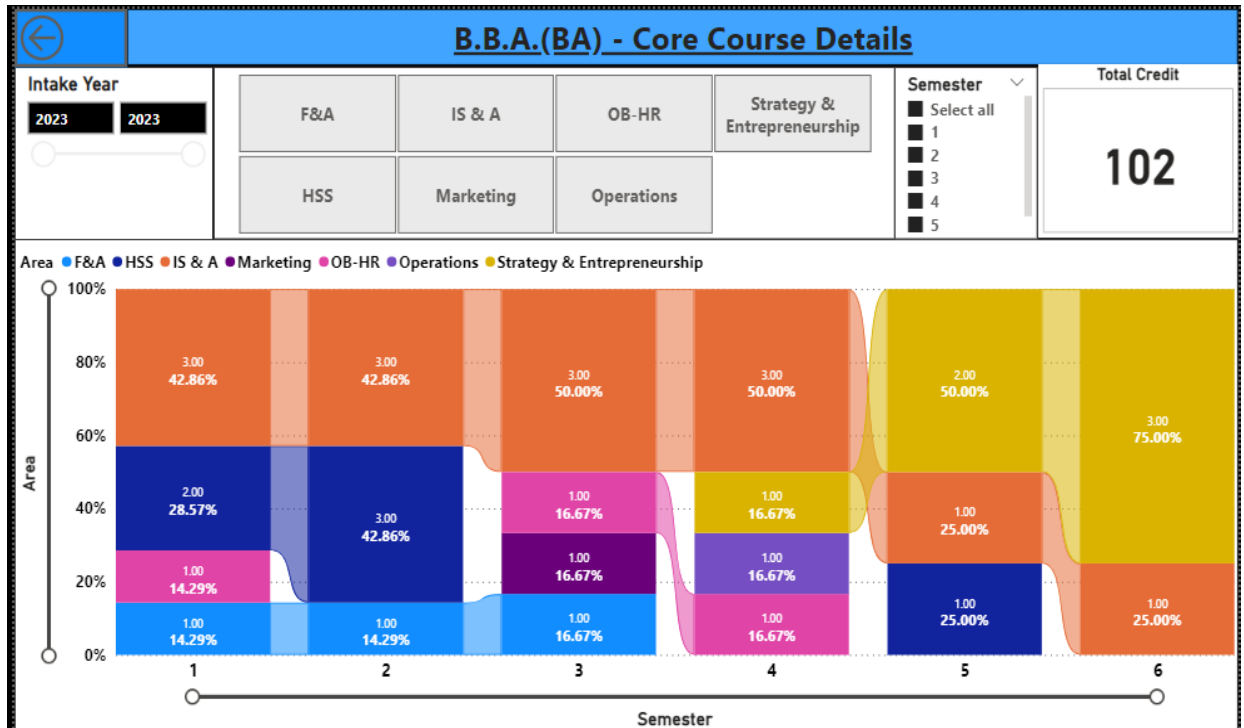
The data for this report was sourced through secondary means, obtained directly from the Administration Department of the JGBS (OP Jindal Global University). The provided dataset comprises raw excel data detailing courses within various programs, including information on semesters and credit allocations. This dataset forms the foundation for our analysis and subsequent development of the Power BI dashboard.

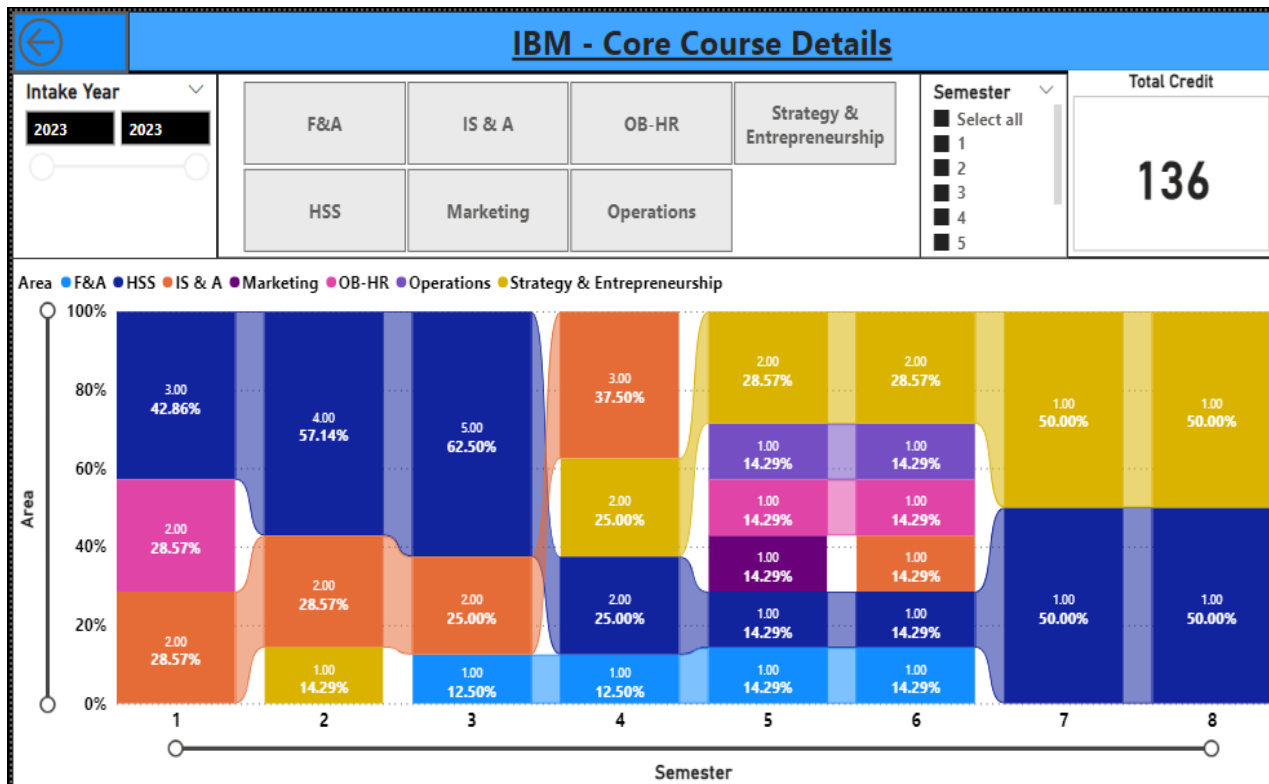
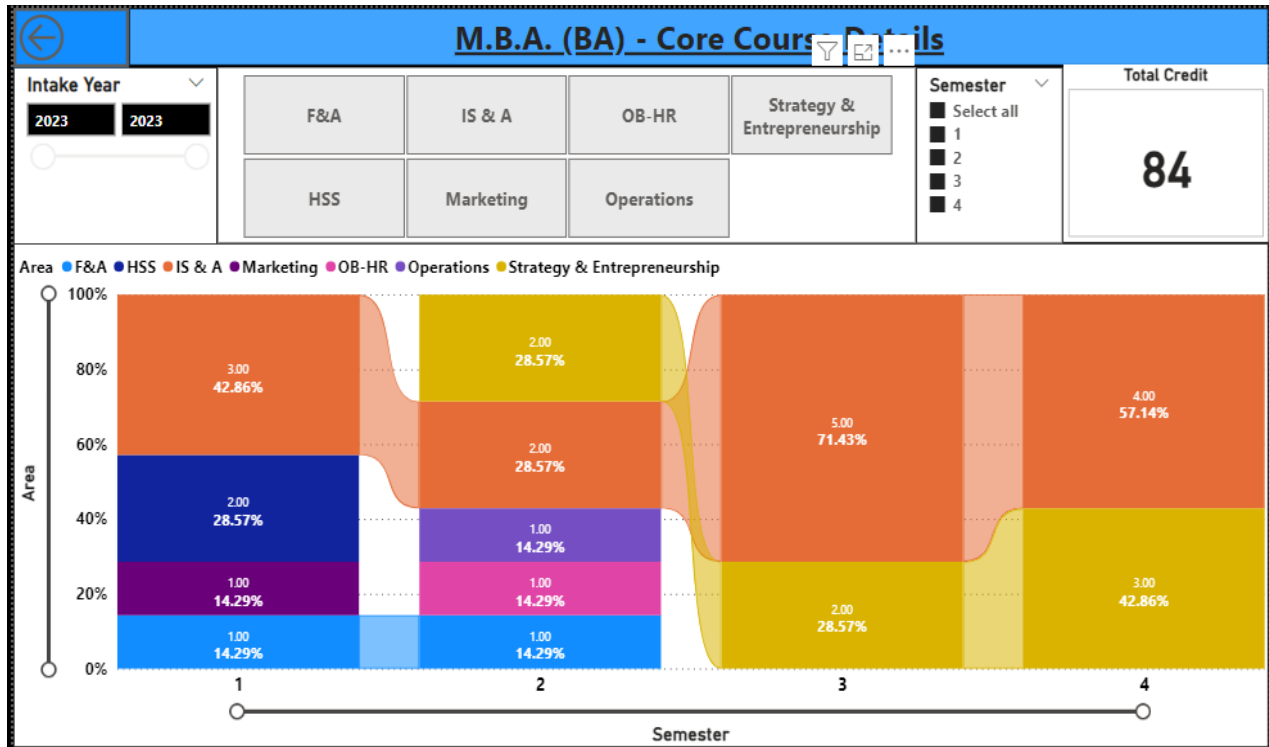
The collaboration with the Administration Department underscores a commitment to transparency and facilitates the transformation of this raw data into a comprehensible visual representation, contributing to a more insightful understanding of the academic structure at JGBS.

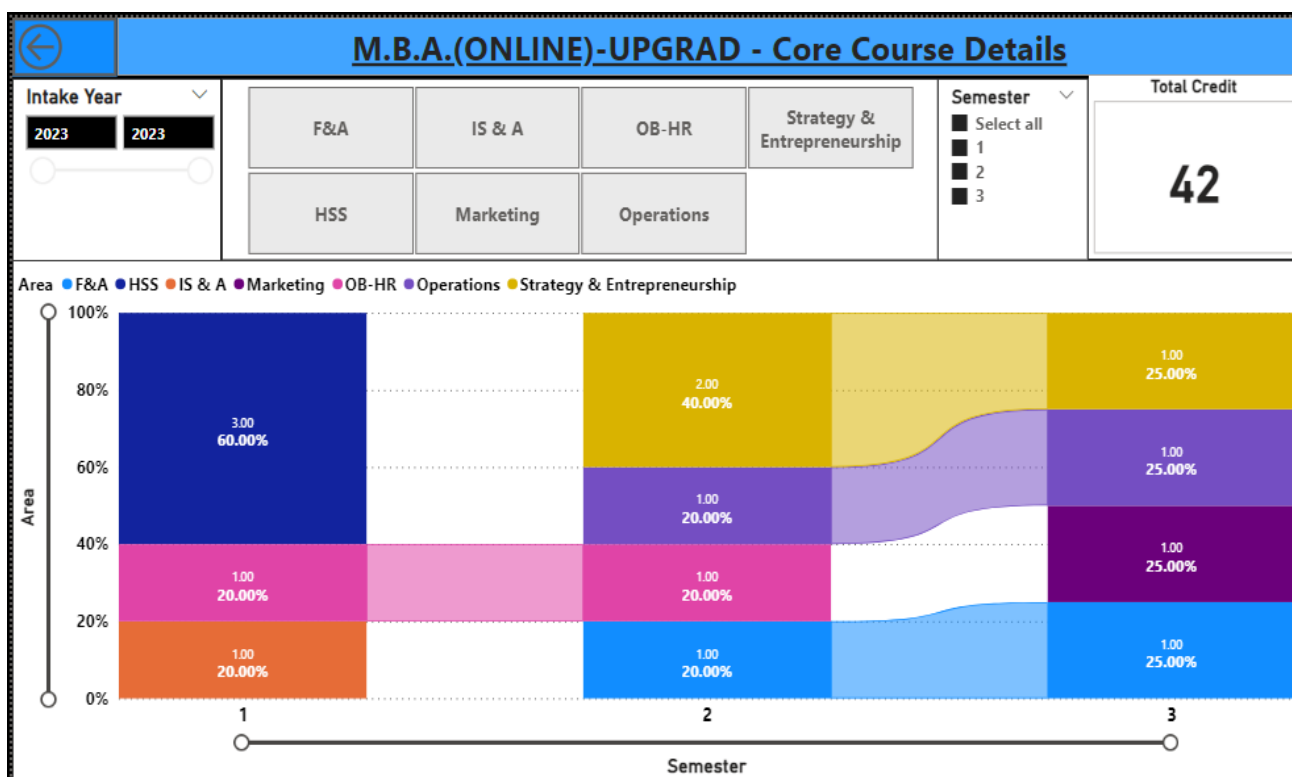
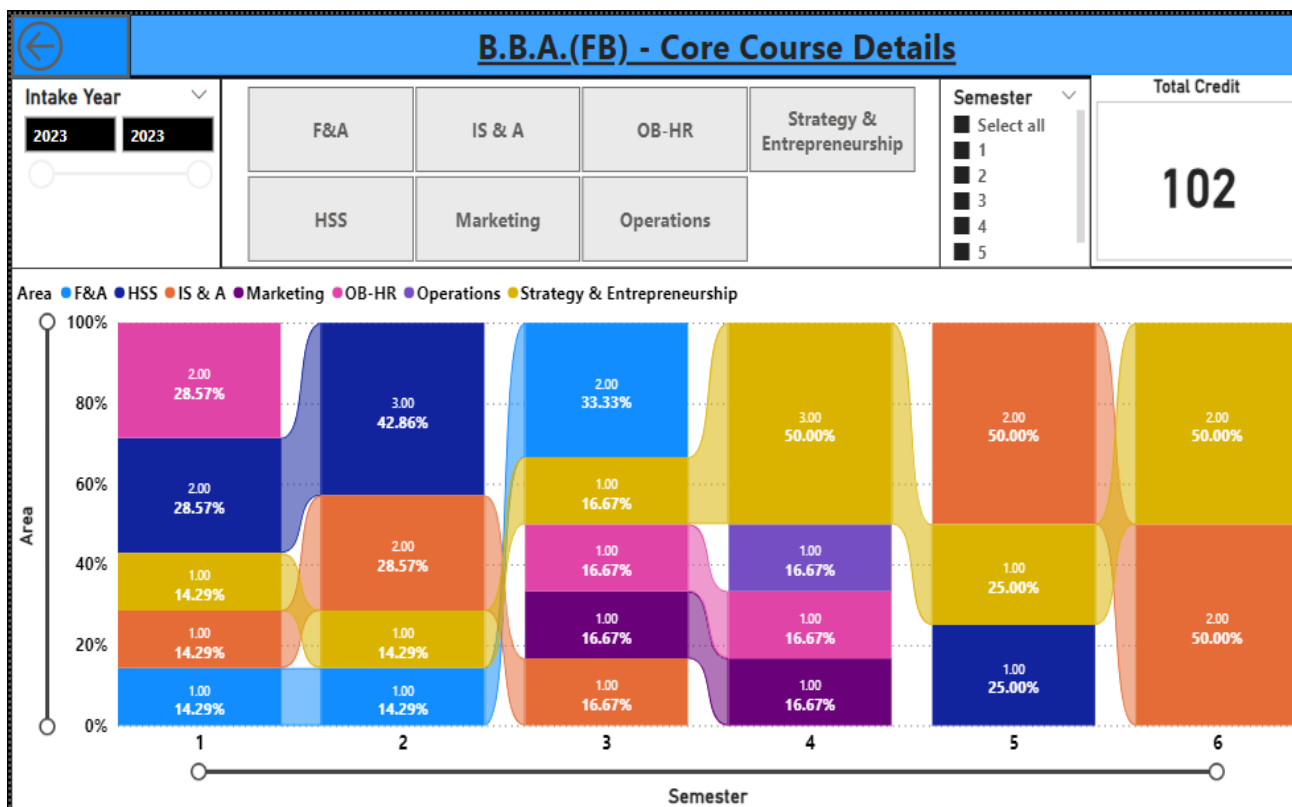
Methods and Materials

The data was obtained in raw form in Excel Sheet that was predominantly cluttered and difficult to comprehend. This data was cleaned and organized within Excel and prepared to take a structured format. This structured data was thereafter imported to Power BI for further analysis and visualizations. Once the data was imported, the stakeholders were consulted to arrive at the type of visualization required by them for easy analysis. After confirming this, we proceeded to opt for the relevant visualization tools to represent the data as per the stakeholder's requirement. Ribbon Chart was chosen for the primary representation due to its inherent ability to depict flow of elements within, in this case it was the flow of courses across semesters. We aimed to represent maximum data in one page with the objective of presenting a one-stop solution to each of the programs offered at JGBS. JGBS has a variety of business management programs. A visual representation was created which included program distribution, core course subjects, course credits, semester wise distribution of courses, and how it's carried forward each semester and other relevant metrics.

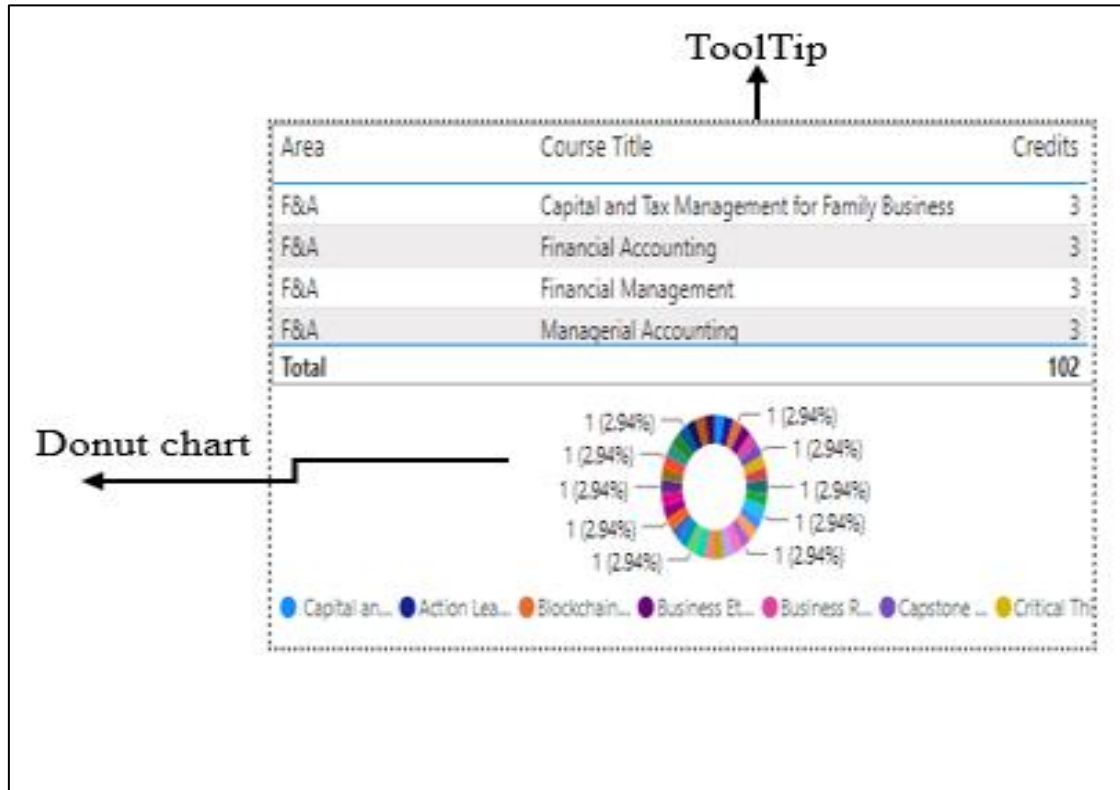








Tooltip for program's course



Types of functions/graphs/tables/cards/etc.

A variety of interactive elements were used to effectively communicate insights and facilitate data exploration for the users of this dashboard. These include slicers, cards, ribbon charts, action buttons, zoom slider, tooltips, pie charts, and data tables.

Visualization

Slicers are interactive filters that allow users to dynamically slice and dice data across multiple visualizations. They can be based on categorical fields like product categories, regions, or time periods. Users can select specific values from slicers to filter data displayed in other visuals on the dashboard, enabling ad-hoc analysis and exploration. In the JGBS dashboard, **tiles were added through slicer** were used to showcase the different areas of study, for example, Finance

and Accounting (F&A), Strategy and Entrepreneurship, Marketing, etc. Moreover, the conditions of intake year and number of semesters were also added through slicer.

Cards are compact visualizations used to display key metrics or KPIs prominently. They can be customized to show single numerical values, such as total sales, revenue, or customer count. Cards can also be formatted to include icons, trend indicators, or conditional formatting to highlight performance against targets or benchmarks. Cards were used to showcase the total credits in a particular area of study within the different programs.

Ribbon charts display horizontal bars with varying widths representing the proportion of each category. Ribbon charts are effective for showing ranking, distribution, or composition of data in a visually appealing manner. Through the dashboard, ribbon charts can clearly show the distribution of the different areas of study in various semesters and the shift of focus in all semesters.

The **zoom slider** enables users to adjust the level of detail or granularity displayed in certain visualizations, such as time series charts or geographical maps. It allows users to focus on specific time periods or geographic regions by zooming in or out dynamically. The zoom slider enhances the user experience by providing greater control over data exploration and analysis. This was applied for semester and area to make the dashboard more detailed and user friendly.

Tooltips provide additional context or details about data points when users hover over them with their mouse cursor. They can display relevant information such as values, labels, or insights associated with specific data points. Tooltips help users understand the data better and uncover insights without cluttering the visualizations with excessive information. **Pie charts** and **data tables** were easily accessible by tooltips. Pie charts are effective for visualizing the composition of a whole and comparing the proportions of different categories. Tables within tooltips provide detailed data when users hover over specific data points in visualizations. They can display additional attributes, metrics, or contextual information related to the selected data point. The various subjects being taught were visualized through pie charts, and the table showed the data in an organized manner along with the credits.

Navigation

Two functions were used for navigation. Firstly, a **back button** was added to provide users with a convenient way to navigate back to a previous state or page within the dashboard. Since nine courses are being covered within the same dashboard, using a back button saves time and makes the process of finding information more efficient. Secondly, **blank queries** were used to perform various data transformation tasks or calculations behind the scenes. The **last refresh button** displayed on the cover page of the JGBS course structure dashboard shows the date and time of the last refresh. This was made by using blank query, tables, measures, and displayed on a card.

Steps for Analysis

The present project was aimed to produce a simplified visualization of a multitude of data that was cumbersome to navigate through earlier. For this very reason, we began with the aim of ensuring that the visualization enables a hassle-free analysis from the user's end.

1. From the user's perspective, it seemed convenient to have one home page dedicated to showing the listed of programs offered by JGBS. This makes it easy for the user to pick a course that they want to analyze further. Additionally, the "Last Refresh Button" conveniently lets the user know that the data they want to analyze is indeed up to date.
2. Upon clicking on a program, the user will be directed to the page entailing the details of the selected program. This page must showcase important details such as the *academic year*, *domains of specialization*, *number of semesters* and *total credits* achieved by a student in the chosen program. Tools such as slicers and cards were used for achieving the same.
3. Furthermore, the ribbon chart displays the break-up in percentage of the various domains across the semesters. The X-Axis of the ribbon chart denotes the semester number, and the Y-Axis denotes the percentage value for the Area break-up. The connected ribbons allow for a clear visual of the flow of a particular area across each semester enabling the user to analyze the strength of presence of an area throughout the program.
4. It is not tomfoolery to assume that the user would also want to know what the courses are in a semester in a particular area and how many courses are offered in that area. This is where we introduced the use of customized tooltips that give all this information by simply moving the cursor over to an area on the ribbon chart. The tooltip instantaneously shows the name of the course offered along with the number of credits earned by the student for that course in the form of a table in addition to a donut chart that shows what percentage weightage does a particular course hold in the total number of courses offered in that domain. This allows the user to analyze the nature of courses and helps come up with a new course perhaps for the next academic year in that domain.
5. A clear visual of this data enables the user to also analyze and assess the expected knowledge level of a student until a given semester in a particular domain thereby being able to evaluate the impact of introduction of a new course in succeeding semesters.

Results and Discussion

Power BI is an extremely powerful tool that has the capability to convert large data into presentation ready visuals. By using Power BI, a single dashboard was created where all information regarding the program distribution, course subjects, credits, and semester-wise trends at Jindal Global Business School (JGBS) is presented. Through this dashboard, the challenges of displaying complex and large data in a user friendly and efficient way was overcome and now, the dashboard can be used to empower faculty, administrators, and students,

streamlining data analysis, facilitating informed decision-making, and contributing to the continuous improvement of academic programs at JGU.

The dashboard offers a comprehensive overview of the distribution of courses across various departments and areas of study. It reveals the credit allocation for each course, enabling stakeholders to understand the workload associated with different courses. By examining the distribution of credits across semesters, trends in course offerings and academic requirements become apparent. This information is crucial for students in planning their course schedules and for administrators in managing academic resources effectively.

Key Takeaways:

1. In the case of MBA, the core courses across the 4 semesters seems to be most inclined towards IS & Analytics and Strategy and Entrepreneurship domains followed by HSS domain. This could be because Analytics and Strategy is a driving force in all domains today and has become the need of the hour to make well-informed decisions.

Semester	F&A	HSS	IS & A	Marketing	OB-HR	Operations	Strategy & Entrepreneurship	Total
1	1	2	2	1	1	1		8
2	1	1	2		1	1		8
3		1	1					3
4								2
Total	2	4	5	1	2	2	5	21

2. The same trend is observed in the case of BBA (Hons) with the maximum weightage for IS & Analytics and Strategy and Entrepreneurship domains.

Semester	F&A	HSS	IS & A	Marketing	OB-HR	Operations	Strategy & Entrepreneurship	Total
2	1	3	3				1	8
1	1	2	2		2			7
4			1		1	1		6
3	1		1	1	1			5
5		1	1	1				4
6			1			1		4
Total	3	6	9	2	4	2	8	34

3. The BBA – Business Analytics program has rightfully incorporated a judicious number of core courses under the IS & Analytics domain which only adds to solidifying the credibility of the offered program.

Semester	F&A	HSS	IS & A	Marketing	OB-HR	Operations	Strategy & Entrepreneurship	Total
1	1	2	3		1			7
2	1	3	3					7
3	1		3	1	1			6
4			3		1	1		6
5		1	1					4
6			1					4
Total	3	6	14	1	3	1	6	34

4. The BBA – Family Business entails a major chunk in Strategy & Entrepreneurship which contributes to the idea of the program in the first place.

Semester	F&A	HSS	IS & A	Marketing	OB-HR	Operations	Strategy & Entrepreneurship	Total
1	1	2	1			2		7
2	1	3	2					7
3	2		1	1	1			6
4				1	1	1		6
5		1	2					4
6			2					4
Total	4	6	8	2	4	1		34

5. The BBA – Financial Management program has the Finance and Accounting domain dominating with almost in-par competition from the IS & Analytics Domain. This simply reiterates the need for being well-versed in Analytics in today's world and makes the program offered at JGBS competitive.

Semester	F&A	HSS	IS & A	Marketing	OB-HR	Operations	Strategy & Entrepreneurship	Total
1	2	2	2			1		7
2	2	3	2					7
3	3		1	1	1			6
4	2		1		1	1		6
5		1	1					4
6			1					4
Total	9	6	8	1	3	1		34

6. The Integrated BBA-MBA (IBM) program comprises of a major chunk of Humanities and Social Science domain courses within the core courses curriculum followed by IS & Analytics. This program could still offer a better range under the various domains.

Semester	F&A	HSS	IS & A	Marketing	OB-HR	Operations	Strategy & Entrepreneurship	Total
3	1	5	2					8
4	1	2	3					8
1		3	2		2			7
2		4	2					7
5	1	1		1	1	1		7
6	1	1	1		1	1		7
7		1						2
8		1						2
Total	4	18	10	1	4	2		48

7. The One-Year Global MBA program at JGBS is specialized in Business Analytics and does justice to the same by offering core courses that are predominantly within the IS & Analytics domain.

Semester	F&A	HSS	IS & A	Marketing	OB-HR	Operations	Strategy & Entrepreneurship	Total
1	1	1	3					5
2			3		1	1		5
3	1		3	1				5
4			3		1			5
5			3					5
6	1		1	1				4
Total	3	1	16	2	2	1		29

Implications

Several implications can be drawn from the insights derived through the dashboard. Regularly reviewing and updating the curriculum is crucial to keep it in line with industry trends and student interests. This means assessing existing courses, spotting emerging areas of study, and integrating new developments in business education. Additionally, strategic planning of course offerings is essential to ensure a balanced distribution of credits across semesters and areas of study. This helps maximize student engagement and satisfaction while making efficient use of

academic resources. Lastly, administration tasks can become more streamlined resulting in fewer mistakes and quick decision making.

Drawbacks

While the Power BI dashboard showcasing the courses offered at Jindal Global Business School offers valuable insights, it also presents some drawbacks. These include potential limitations in interactivity (i.e., appeal of visualization through the charts used is highly subjective to the user and hence if a user prefers a different chart, then they have to make the changes in the Power BI Report), complexity of visualization (the user must be well-versed with interpretations of different kinds of chart which if in case not present, proper training must be provided from the organization's end. The Ribbon Chart used here is an advanced chart that requires awareness from the user's end to interpret), data accuracy (data in the excel must be accurate and up to date for reliability of the visualization and to avoid confusion). Accessibility issues, security and privacy concerns also pose challenges.

Addressing these drawbacks requires careful consideration of dashboard design, data quality assurance, user feedback mechanisms, and adherence to best practices in data visualization and accessibility standards to ensure the dashboard's effectiveness and usability for decision-making and performance monitoring.

Recommendations

To address the implications identified, several recommendations can be proposed. Firstly, Jindal Global Business School (JGBS) needs to create a structured and standardized procedure for the development of a new curriculum and the consistent review of this curriculum while also inviting collaboration amongst various stakeholders like the academic staff, business professionals, and student representatives. This procedure will help corroborate the flexibility and adaptability of the curriculum. Secondly, the underrepresented areas of the university can be addressed by including targeted marketing and promotions. Lastly, the university may draw in a diverse student population and cultivate a lively academic community by actively and efficiently communicating the value proposition of these courses. Moreover, this dashboard can be implemented in other schools like Jindal Global Law School and Jindal School of Liberal Arts and Humanities to streamline the process. Areas like faculty profiles, exams, admissions, transportation, and food committee can also benefit from this dashboard.

Limitations and Scope of study

It is important to address the limitations of this dashboard. The focus of the dashboard is only on the course offers, credits, intake years, and study topics. Even though this may help in creating informed decisions and identifying useful trends, the dashboard misses out on other crucial facets of the student experience, like engagement, satisfaction, and career results. Moreover, the qualitative aspects of student life and academia are overlooked because of the sole use of

quantitative data. These limitations help in improving the scope of this study. For example, the dashboard can offer a more thorough perspective of course offerings, curriculum frameworks, and academic trends across the university by combining data from other schools like Jindal Global Law School and Jindal School of Liberal Arts and Humanities. This comparison analysis can assist in identifying each school's areas of strength and progress as well as chances for cooperation and knowledge exchange. Additionally, more qualitative areas can be incorporated which enhances the dashboard's ability to portray information in a cohesive manner and help in critical decision making. This process also aids in understanding the current ranking of Jindal Global Business School, comparing it to its competitors, and stressing areas of competitive advantage and potential areas for additional investment or strategic efforts to strengthen or preserve its position. Therefore, with an increased scope of study, all the relevant stakeholders can make well-informed decisions to promote academic programme excellence and ongoing improvement by gaining insightful knowledge of the larger educational environment.

Conclusion

Therefore, the motivation behind creating a dashboard using power BI was to give a holistic rundown of the course curriculum at JGU's Jindal Global Business School (JGBS) using information from the administration office of the university. By using various visualization tools, stakeholders can be empowered to identify various trends and gain important insights that can be used to implement continuous improvements in the course curriculum leading to a more refined course that benefits the students. Choosing and analyzing variables like course offerings, programme distribution, course subjects, credits, and trends over the course of the semester helped in locating patterns in the curriculum that work while also finding areas where improvement is needed.

By incorporating this dashboard in the universities administration, the universities commitment to transparency and active involvement in the constant enhancement of the course can be highlighted. Moreover, it created a culture of collaboration and efficiency amongst the users and planners of this dashboard. This dashboard also acts as a gateway to the world of data analytics. The potential of using data to make critical decisions is incredibly vast and its implementation in the education system is reinforced through this project.

References

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2. Guy in a Cube. (2020, June 16). *How do you get a REFRESH DATE in your Power BI report???* [Video]. YouTube. <https://www.youtube.com/watch?v=oN6mOmEruOQ>

